EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).
- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also

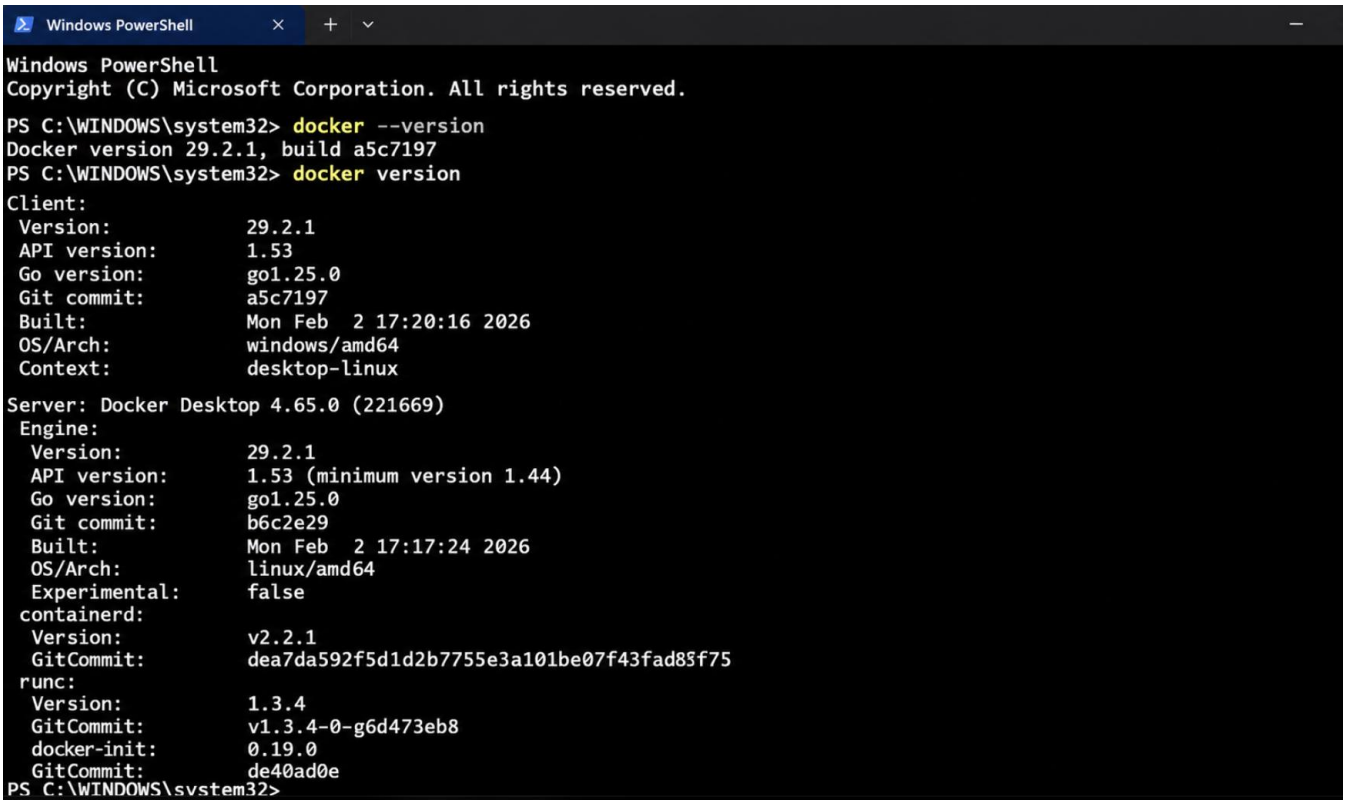
supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

1 - Pre-requisite: Install Docker Desktop

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:



Also make sure the WSL2 feature is enabled:

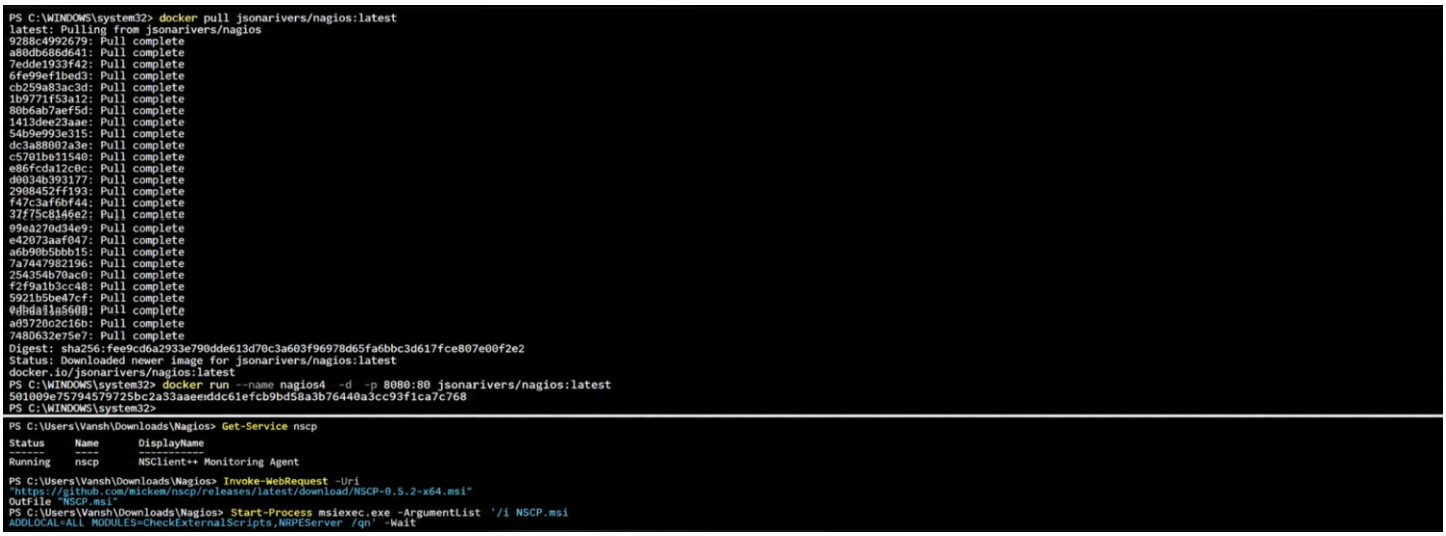
```
PS C:\Users\Vansh\Downloads\Nagios> .\NSCP-0.5.2-x64.msi /quiet /norestart
Installing NSClient++ (x64) ...
A later version will be installed already exists. No need to uninstall.
Installation completed successfully.
PS C:\Users\Vansh\Downloads\Nagios> nscp --version
NSClient++ 0.5.2.62 2024-01-28
Build: master-0.5.2.62
Module directory: C:\Program Files\NSClient++\modules
Config file: C:\Program Files\NSClient++\nsclient.ini
PS C:\Users\Vansh\Downloads\Nagios> nscp service
STATE      SERVICE      DISPLAY NAME                VERSION
-----      -------      ------------                -------
running    nscp         NSClient++ Monitoring Agent   0.5.2.62

NSClient++ is running. Use nscp settings --help for commands.
For more information and the complete documentation, visit:
https://nsclient.org

PS C:\Users\Vansh\Downloads\Nagios>
```

2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.



Verify the container is up and running:

3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:



Enter and confirm the password when prompted (e.g. 123456).

4 - Access Nagios Web Interface

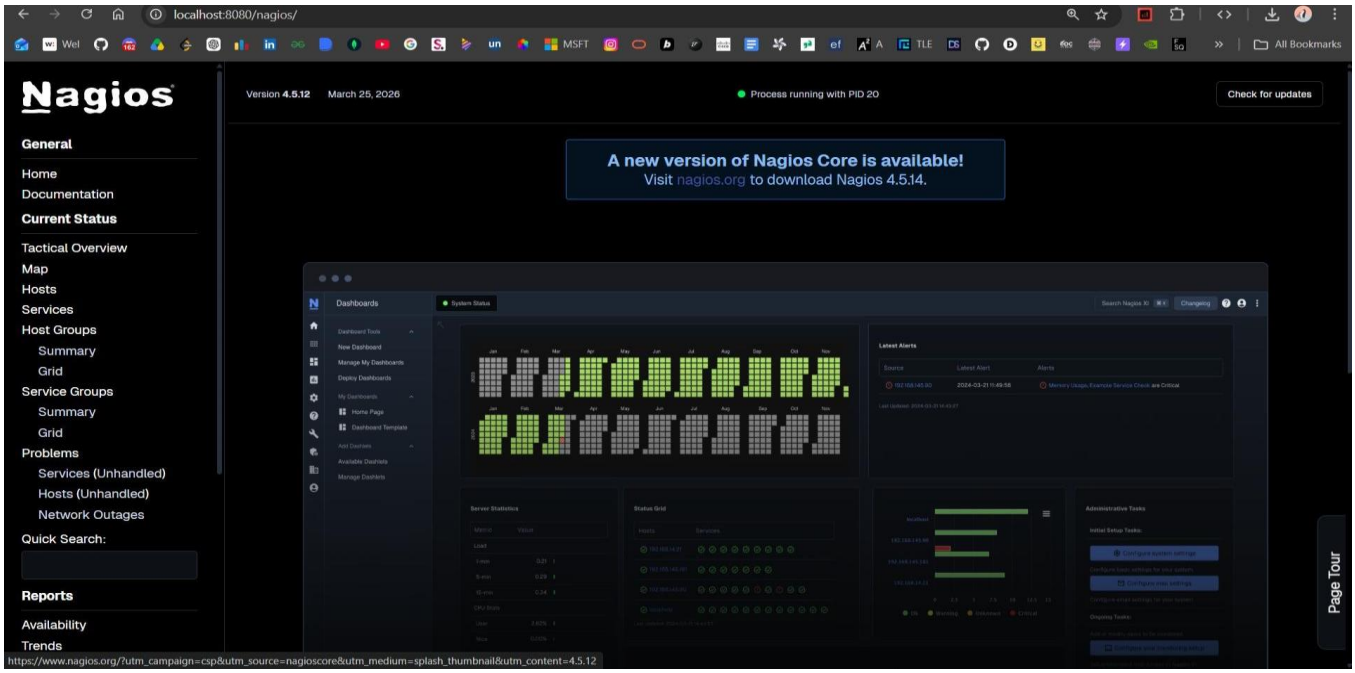
Access the Nagios web interface from a browser on the Windows host:

http://localhost:8080/nagios/

Prompting for Authentication Password -

username: nagiosadmin

Password: 123456 (which you entered while configuration)



5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

```
PS C:\Users\Vansh\Downloads\Nagios> Get-Service nscp

Status   Name               DisplayName
------   ----               -----------
Running  nscp               NSClient++ Monitoring Agent

PS C:\Users\Vansh\Downloads\Nagios>
```

6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

File Edit View

```
# If you want to fill this file with all available options run the following command:
#   nscp settings --generate --add-defaults --load-all
# If you want to activate a module and bring in all its options use:
#   nscp settings --activate-module <MODULE NAME> --add-defaults
# For details run: nscp settings --help


; in flight - TODO
[/modules]

; Load on startup - If the module should be loaded on startup
NRPEServer = enabled

; Load on startup - If the module should be loaded on startup
WEBServer = enabled

; Load on startup - If the module should be loaded on startup
CheckDisk = enabled

; Load on startup - If the module should be loaded on startup
CheckSystem = enabled

; Load on startup - If the module should be loaded on startup
CheckEventLog = enabled

; Load on startup - If the module should be loaded on startup
CheckHelpers = enabled

; Load on startup - If the module should be loaded on startup
CheckExternalScripts = enabled

; Load on startup - If the module should be loaded on startup
CheckNSCP = enabled


; in flight - TODO
[/settings/NRPE/server]

verify mode = none

tls version = tlsv1.2+

insecure = true

allowed ciphers = ADH


; in flight - TODO
[/settings/default]

allowed hosts = 127.0.0.1,172.17.0.2
```

Ln 31, Col 1 | 1,269 characters | Plain text

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

After saving the file, restart the NSClient++ service so the change takes effect:

7 - Verify Connectivity from Nagios

From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

```
PS C:\Users\Vansh> Invoke-WebRequest -Uri
"https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi"
OutFile "NSCP.msi"
PS C:\Users\Vansh> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi
ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait
```

8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

```
root@501099e75794:/# printf '%s\n' 'define host {' '    use                     windows-server' '    host_name               Windows_Host_001' '    alias                   Windows Host 001' '    address
       192.168.0.147' '    register                1' '}' '' 'define service {' '    host_name               Windows_Host_001' '    service_description     PING' '    check_command           check_ping!100.0,20
%!500.0,60%' '    max_check_attempts      2' '    check_interval          2' '    retry_interval          2' '    check_period            24x7' '    check_freshness         1' '    contact_groups          admins
' '    notification_interval   2' '    notification_period     24x7' '    notifications_enabled   1' '    register                1' '}' > /opt/nagios/etc/servers/MyWindowsHost001.cfg
root@501099e75794:/# cat /opt/nagios/etc/servers/MyWindowsHost001.cfg
define host {
    use                     windows-server
    host_name               Windows_Host_001
    alias                   Windows Host 001
    address                 192.168.0.147
    register                1
}

define service {
    host_name               Windows_Host_001
    service_description     PING
    check_command           check_ping!100.0,20%!500.0,60%
    max_check_attempts      2
    check_interval          2
    retry_interval          2
    check_period            24x7
    check_freshness         1
    contact_groups          admins
    notification_interval   2
    notification_period     24x7
    notifications_enabled   1
    register                1
```

```
define host {
   use             windows-server
host_name       Windows_Host_001
alias           Windows Host 001    address
192.168.1.50
   register        1
}
define service{     host_name               Windows_Host_001
service_description         PING            check_command
check_ping!100.0,20%!500.0,60%      max_check_attempts
2   check_interval     2   retry_interval     2   check_period
24x7     check_freshness     1     contact_groups     admins
notification_interval 2   notification_period   24x7
   notifications_enabled  1
   register            1
}
```

Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

```
root@501099e75794:/# /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg

Nagios Core 4.5.12
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-03-25
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
       Checked 8 services.
Warning: Host 'Windows_Host_001' has no default contacts or contactgroups defined!
       Checked 2 hosts.
       Checked 1 host groups.
       Checked 0 service groups.
       Checked 1 contacts.
       Checked 1 contact groups.
       Checked 23 commands.
       Checked 5 time periods.
       Checked 0 host escalations.
       Checked 0 service escalations.
Checking for circular paths...
       Checked 2 hosts
       Checked 0 service dependencies
       Checked 0 host dependencies
       Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 1
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
root@501099e75794:/#
```

9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

